

SentiAvatar: Towards Expressive and Interactive Digital Humans

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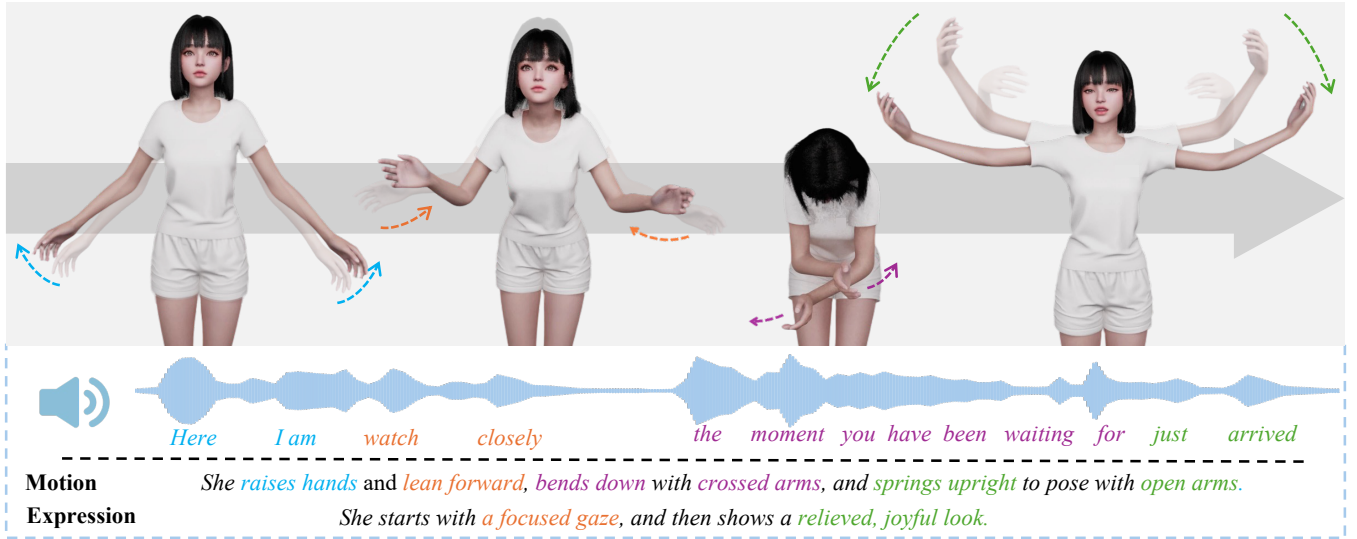


Figure 1: Our SentiAvatar generates high-quality 3D human motion and expression, which are semantically aligned and frame-level synchronized. Texts and arrows of the same color indicate the same time step.

Abstract

We present **SentiAvatar**, a framework for building expressive interactive 3D digital humans, and use it to create **SuSu**, a virtual character that speaks, gestures, and emotes in real time. Achieving such a system remains challenging, as it requires jointly addressing three key problems: the lack of large-scale high-quality multimodal

data, robust semantic-to-motion mapping, and fine-grained frame-level motion-prosody synchronization. To solve these problems, first, we build **SuSuInterActs** (21K clips, 37 hours), a dialogue corpus captured via optical motion capture around a single character with synchronized speech, full-body motion, and facial expressions. Second, we pre-train a **Motion Foundation Model** on 200K+ motion sequences, equipping it with rich action priors that go well beyond the conversation. We then propose an audio-aware **plan-then-infill** architecture that decouples sentence-level semantic planning from frame-level prosody-driven interpolation, so that generated motions are both semantically appropriate and rhythmically aligned with speech. Experiments show that SentiAvatar achieves state-of-the-art on both SuSuInterActs (R@1 43.64%, nearly 2× the best baseline) and BEATv2 (FGD 4.941, BC 8.078), producing 6 s of output in 0.3 s with unlimited multi-turn streaming.

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CCS Concepts

• **Computing methodologies** → **Animation**; *Computer vision*;
 • **Information systems** → **Multimedia content creation**; •
Human-centered computing → *Human computer interaction (HCI)*.

Keywords

Interactive 3D Avatar, Human Motion Generation

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1 Introduction

People communicate with more than words. A shrug carries helplessness; a nod signals agreement; a raised eyebrow conveys doubt. These non-verbal behaviors—gestures, postures, facial expressions—are integral to interaction, not mere accessories to speech. Giving 3D digital characters this expressiveness matters across domains: virtual assistants need it to build trust, robots need it to cooperate with humans, and games need it to make characters more vivid. Current research focuses on generating motions from text prompts [17, 19, 21, 40, 48, 65] or producing gestures from speech audio [4, 30, 33, 61, 74]. Few works study how to generate motions that are semantically consistent with dialogue context and intended utterances in interactive scenarios.

Building an interactive digital human that naturally gestures and emotes in real dialogue poses three challenges: (1) *Lack of high-quality data*—full-body motion and facial expressions during conversational interaction. Most text-to-motion datasets [18, 29, 43] only pair action descriptions with motion sequences, without dialogue or interaction context; speech-driven datasets such as BEAT [31] and BEATv2 [30] are built from monologue-style presentations rather than bidirectional conversation. (2) *Insufficient semantic-to-motion alignment*—when motion descriptions are complex or lengthy, generated motions often miss certain actions. (bridging the gap between discrete high-level semantic intents and continuous motion dynamics is difficult. When descriptions are complex, existing models often struggle to faithfully execute all intended actions) (3) *Under-explored prosody alignment in dialogue*—existing co-speech methods are trained on single-speaker presentation data, which differs from the motion patterns that arise in person-to-person dialogue.

We present **SentiAvatar**, as shown in Figure 1, a framework for building expressive interactive 3D digital humans, and use it to create **SuSu**, a virtual character that speaks, gestures, and emotes in real time. We tackle all three challenges in a unified pipeline. For *data*, we design multi-scenario dialogue scripts around SuSu with behavior annotations and have professional actors perform them in an optical motion-capture studio, capturing synchronized speech, full-body motion, and facial expressions to form **SuSuInterActs** (21K clips, 37 hours). To improve motion generalization and diversity, we pre-train a **Motion Foundation Model** on 200K+

sequences aggregated from public corpora and the Tencent Hunyuan Motion Model, giving SuSu rich action priors that go well beyond the conversational domain. With data and motion knowledge in place, we propose a new framework, **SentiAvatar**, that achieves both semantic alignment between text prompts and generated motions and rhythmic alignment between generated motions and speech. Our audio-aware **plan-then-infill** architecture decouples these two objectives: the fine-tuned foundation model takes behavior labels and audio tokens as input and generates semantically aligned sparse keyframes; an Infill Transformer then fills in the intermediate frames conditioned on boundary keyframes and frame-level speech features, producing motions that are both semantically appropriate and rhythmically aligned. The resulting system achieves state-of-the-art on both SuSuInterActs and BEATv2 [30], producing 6 s of output in 0.3 s with unlimited multi-turn streaming.

Our contributions are three-fold: **First**, we open-source an interactive 3D digital human framework: SuSu speaks, gestures, and emotes in real-time dialogue. **Second**, we construct SuSuInterActs, a 21K-clip, 37-hour multimodal dialogue corpus with synchronized speech, behavior-annotated text, full-body motion, and facial expressions. **Third**, we propose the plan-then-infill architecture and a Motion Foundation Model pre-trained on 200K+ sequences, achieving state-of-the-art on both SuSuInterActs (R@1 43.64%, nearly 2× the best baseline) and BEATv2 (FGD 5.301→4.941, BC 7.971→8.078, improving the best prior results on both metrics).

2 Related Work

2.1 Text-driven Human Motion Generation

Text-driven motion generation has progressed from VAE models [19, 40, 47] to diffusion methods [12, 20, 48, 50, 64, 67], real-time control [37, 70], and discrete-token modeling [9, 16, 41, 42, 52, 62, 63, 66]. Motion LLMs [22, 38, 53, 54, 72, 73] improve semantic reasoning but still generate from static text without speech prosody. SentiAvatar instead plans sparse dialogue-conditioned keyframes that a prosody-aware Infill Transformer densifies.

2.2 Speech-driven Gesture and Expression Generation

Co-speech methods learn from video [2, 14] and improve synchronization by hierarchical, rhythm-aware, diffusion, discrete, and contrastive objectives [3–5, 10, 32, 33, 58, 74]. Holistic models cover body, hands, and face [8, 30, 59, 60], while recent systems explore conversation [36, 44, 57, 71] and text guidance [6, 7, 39, 69]. Most still treat gestures as responses to speech rhythm. SentiAvatar instead separates sentence-level semantic planning from prosodic infilling and jointly generates body motion and facial expressions.

2.3 Motion Representation, Foundation Models, and Datasets

VQ-VAE representations [16, 58, 66] connect our LLM planner and Infill Transformer through discrete motion tokens. Large-scale pre-training is emerging in Kimodo, GENMO, ViMoGen, and ViMoRAG [24, 28, 45, 55], motivating our 200K-sequence foundation model. Existing datasets cover text-motion [18, 29, 43, 68], co-speech [11, 30, 31], dance [25, 26], and large-scale or interactive motion [1, 13, 15, 23, 27, 34, 35, 51]. Most are non-conversational

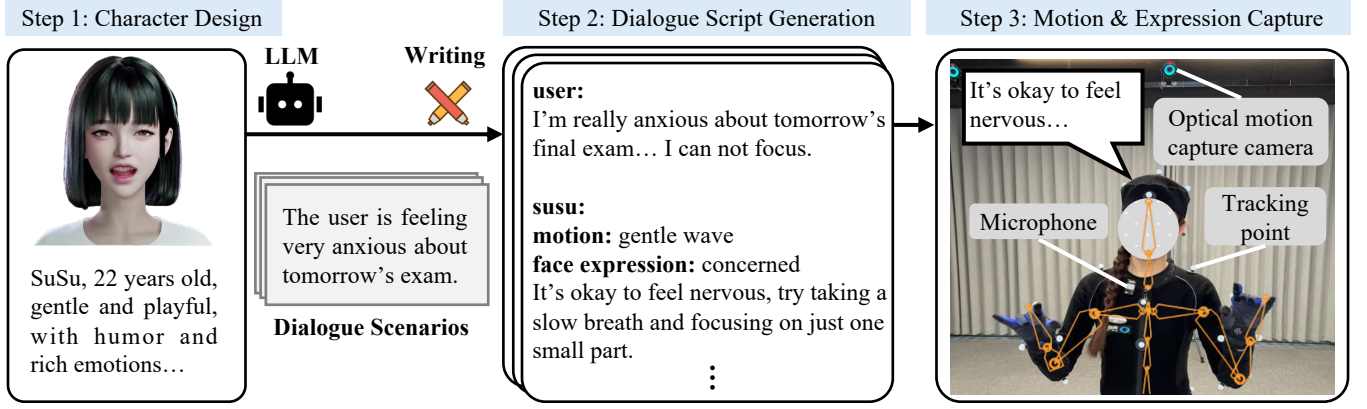


Figure 2: Overview of SuSuInterActs data pipeline. (1) Character design for a consistent persona. (2) LLM script generation with explicit motion/expression tags. (3) Multimodal capture of synchronized speech, full-body motion, and facial expressions.

or lack synchronized faces and semantic behavior labels. SuSuInterActs adds 21K clips (37 hours) of Chinese dialogue with synchronized speech, full-body motion, facial expressions, and per-utterance behavior labels.

3 Constructing SuSuInterActs Dataset

We construct **SuSuInterActs**, a multimodal Chinese conversation dataset centered on a single virtual character. Each sample pairs temporally aligned speech audio, dialogue text, full-body motion, and facial expressions. Formally:

$$\mathcal{D} = \{(C_i, S_i)\}_{i=1}^N, \quad (1)$$

where C_i is an interaction scenario and S_i the corresponding dialogue session. Since the dataset focuses on a single character, the role definition is shared across all samples.

As shown in Figure 2, the collection proceeds in three stages: character design, script generation, and performance capture.

Character Design. SuSu is a 22-year-old virtual character with a warm yet playful personality. Her emotional range—curiosity, encouragement, teasing, bashfulness—ensures diverse motion and expression patterns in the captured data. The full character profile is provided in the supplementary materials.

Dialogue Script Generation. We design dialogue scenarios spanning everyday topics—casual chat, storytelling, emotional support, playful banter—to elicit diverse emotional states and interaction dynamics. We then prompt a large language model to generate a multi-turn dialogue script based on the given scenario C_i . Each utterance turn is annotated as:

$$\tau_{i,t} = (\mathbf{e}_{i,t}, \mathbf{a}_{i,t}, \mathbf{u}_{i,t}), \quad (2)$$

where $\mathbf{e}_{i,t}$ is a facial expression label (e.g., “raised eyebrows”), $\mathbf{a}_{i,t}$ a body action label (e.g., “tilts head”), and $\mathbf{u}_{i,t}$ the spoken text. All labels use concrete, performable phrases formatted as inline tags. Each script is manually reviewed to filter impractical actions.

Motion & Expression Capture. Professional actors study the character profile and script before each session; light improvisation consistent with SuSu’s personality is permitted. Full-body motion including hand gestures is recorded via optical motion capture.

Statistic	Value
# Total Samples	21,133
Total Duration	36.9 hours
Avg. Duration per Sample	6.3 s
Avg. Characters per Utterance	18.7
# Samples w/ Non-default Expression	9,412
# Samples w/ Non-default Action	14,278
# Samples w/ Any Non-default Label	18,696
Total Motion Frames	2,656,484
Avg. Frames per Sample	125.7
# Samples w/ Facial Data	12,367
Frame Rate	20 FPS
Motion Representation	6D rotation, 63 joints
Face Representation	51-dim ARKit blendshape

Table 1: Statistics of the SuSuInterActs dataset.

Facial expressions are captured with iPhone ARKit as 51-dim blendshape coefficients. Speech audio is also recorded. Each turn yields:

$$\mathbf{f}_{i,t} = (\mathbf{m}_{i,t}, \mathbf{b}_{i,t}), \quad (3)$$

where $\mathbf{m}_{i,t} \in \mathbb{R}^{L \times 393}$ is the full-body pose sequence (6D rotation for 63 joints: 25 body + 19×2 hand) and $\mathbf{b}_{i,t} \in \mathbb{R}^{L \times 51}$ is the blendshape sequence. Raw motion is retargeted onto SuSu’s skeleton, and all modalities are temporally aligned to 20 FPS after quality filtering.

Dataset Analysis. Table 1 summarizes the dataset statistics. SuSuInterActs contains 21K samples totaling 37 hours, with an average of 6.3 s and 18.7 Chinese characters per turn. Every sample carries expression and action labels; among them, 14K have non-default actions and 9K have non-default expressions. Facial blendshape data is available for 12K samples. SuSuInterActs is the first dataset that combines multi-turn dialogue, role-conditioned interactions, speech audio, full-body motion with hands, and facial expressions in a single corpus.

Dataset Comparison. Table 2 compares SuSuInterActs with representative motion and dialogue datasets. Co-speech datasets such as BEAT [31] provide audio-motion pairs but lack multi-turn dialogue. Text-to-motion datasets such as HumanML3D [18] and

Dataset	#Samples	Multi-turn	Role-playing Dialogue	Action Description	Audio	Face	Body Motion	Hands Motion
HumanML3D [18]	16k	✗	✗	✓	✗	✗	✓	✗
Motion-X [29]	95k	✗	✗	✓	✗	✗	✓	✓
BEAT [31]	2.5k	✗	✗	✓	✓	✓	✓	✗
SnapMoGen [15]	20k	✓	✗	✓	✗	✗	✓	✗
InterX [56]	31k	✗	✗	✓	✗	✗	✓	✓
InterHuman [27]	6k	✗	✗	✓	✗	✗	✓	✗
CharacterEval [49]	11k	✓	✓	✓	✗	✗	✗	✗
SuSuInterActs (Ours)	21k	✓	✓	✓	✓	✓	✓	✓

Table 2: Comparison between SuSuInterActs and existing motion and role-playing dialogue datasets. SuSuInterActs uniquely combines multi-turn dialogue, role-conditioned interactions, and synchronized multimodal signals including text, audio, facial expressions, and full-body motion with hands.

Motion-X [29] pair action descriptions with body motion, yet contain no audio or conversational grounding. Interaction datasets such as InterX [56] and InterHuman [27] capture multi-person motion without speech or role conditioning. Role-playing dialogue datasets such as CharacterEval [49] support multi-turn role-based dialogue but are text-only. SuSuInterActs is the first to unify all these dimensions—multi-turn dialogue, role conditioning, speech audio, full-body motion with hands, and facial expressions—in a single corpus. Focusing on one character further yields consistent behavioral patterns, benefiting character-specific generation.

4 Method

4.1 System Overview

Dialogue-driven motion generation requires alignment at two temporal scales: sentence-level *semantic alignment* maps dialogue intent to action types (e.g., “shrug,” “nod”), while frame-level *prosody alignment* synchronizes motion dynamics with speech rhythm. SentiAvatar decomposes generation into two stages (Figure 3). In Stage I, an LLM-based *Motion planner* takes behavior labels and sparse audio tokens as input and produces keyframe motion tokens at interval t . In Stage II, a lightweight *Audio-aware Infill Transformer* fills the $t-1$ intermediate frames conditioned on dense audio features. R-VQVAE discrete tokens serve as the unified interface between the two stages. For facial expressions, a separate Face Infill Transformer generates face tokens directly from audio. Since HuBERT features already encode rich linguistic and prosodic semantics, and facial movements are tightly coupled with speech, no additional LLM planning is needed for this pathway.

The following subsections describe discrete representation (§4.2), semantic planning (§4.3), motion infilling (§4.4), face generation (§4.5), motion foundation model (§4.6), and inference pipeline (§4.7).

4.2 Discrete Representation

Residual Motion Tokenizer. We adopt a Residual VQ-VAE (R-VQVAE) [16] to convert continuous joint rotations into discrete tokens. A 1D convolutional encoder downsamples temporally by 2×; 4-layer residual quantization with codebook size 512 per layer produces token groups (r^1, r^2, r^3, r^4) per time step. Layer-specific offsets yield a unified vocabulary of $512 \times 4 = 2,048$ IDs:

$$\hat{r}^k = r^k + 512 \times (k - 1), \quad k \in \{1, 2, 3, 4\}. \quad (4)$$

The decoder reconstructs continuous motion from predicted tokens. Following previous works [16], the R-VQVAE is trained by combines reconstruction, velocity, commitment, and position losses.

Audio Representation. We extract layer-9 features from a Chinese HuBERT model at 50 FPS, downsampled to 20 FPS. Two forms are used: (1) *discrete tokens* via K-means clustering, subsampled at the keyframe interval t for the LLM; (2) *continuous 768-dim feature vectors* at 20 FPS for the Infill Transformers.

4.3 LLM-based Semantic Planning

Generating motion from dialogue requires understanding high-level intent: the model must know *what* action to perform before deciding *how* to perform it frame by frame. We cast this as a planning problem: an LLM reads the motion label and coarse audio context, then outputs a sparse sequence of keyframe motion tokens that sketch the intended action trajectory. By operating at the keyframe level, the LLM focuses on semantic correctness—selecting the right action types and their temporal layout—without being burdened by frame-level dynamics. This sparse plan also provides strong boundary constraints for the subsequent infilling stage.

Task Formulation. Given a motion label T (e.g., shrug helplessly) and sparse audio tokens $\{a_1, a_{1+t}, a_{1+2t}, \dots\}$ sampled at interval t , the LLM generates sparse keyframe motion tokens:

$$\text{Input: } T \oplus [a_1][a_{1+t}][a_{1+2t}] \cdots, \quad (5)$$

$$\text{Output: } [S_t][r_1][r_{1+t}][r_{1+2t}] \cdots, \quad (6)$$

where $[S_t]$ is the keyframe step, and each $r_i = (r_i^1, r_i^2, r_i^3, r_i^4)$ is a 4-token residual group. The sparse audio tokens provide coarse temporal context—overall utterance rhythm, pause locations, emphasis patterns—that helps the LLM place keyframes at appropriate moments even before dense prosody alignment.

Training. The model is initialized from the Motion Foundation Model (§4.6) and fine-tuned on SuSuInterActs via full-parameter SFT. Pre-trained motion priors provide a strong initialization. A *continuation mode* supports seamless multi-turn generation by prepending the last two keyframe pairs from the previous utterance as context (details in the supplementary materials).

4.4 Audio-conditioned Motion Infilling

The LLM-based planner produces keyframes that capture *what* actions to perform, but the gaps between keyframes lack fine-grained dynamics—the subtle accelerations, hesitations, and rhythmic gestures that make motion look natural and speech-synchronized. The

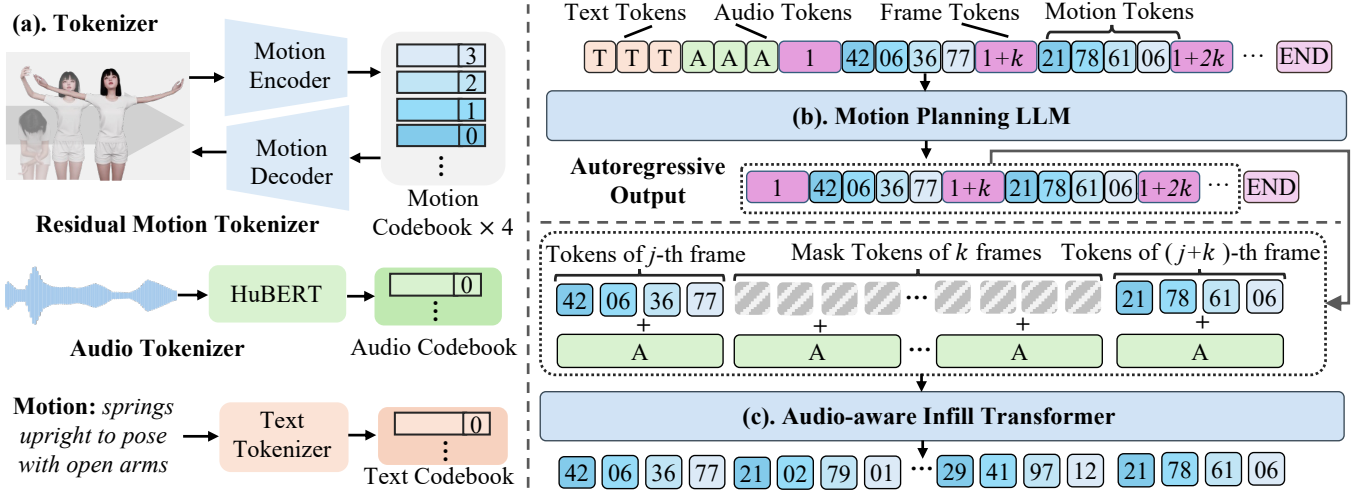


Figure 3: Overview of SentiAvatar. (a) Multi-modal inputs are quantized into tokens via encoders. The body pathway employs a hierarchical generation: (b) LLM planner predicts sparse keyframe tokens to capture high-level dialogue content, followed by (c) an audio-aware Infill Transformer for dense, prosody-driven interpolation to ensure fine-grained temporal synchronization.

Audio-aware Infill Transformer addresses this by conditioning on dense, frame-level audio features to fill intermediate frames, injecting prosody-driven nuance into the coarse plan.

For a sliding window of $t+1$ frames, the two boundary keyframes are known and the $t-1$ interior frames are masked:

$$\text{Input: } [r_i] \underbrace{[M] \cdots [M]}_{t-1} [r_{i+t}] + h_{i:t+t}, \quad (7)$$

where $h \in \mathbb{R}^{(t+1) \times 768}$ are frame-level HuBERT features. The boundary keyframes anchor the motion trajectory while the audio features guide the dynamics—stressed syllables can trigger sharper movements, pauses can induce holds. The model is a Transformer Encoder (8 layers, 16 heads, dim 512, ~38.5M parameters). Audio features are projected via a two-layer MLP and fused with token embeddings by element-wise addition. Learnable positional encodings distinguish temporal positions within the window.

Training. Within the $t-1$ interior frames, a random subset of tokens is masked per sample; unmasked tokens have a 10% probability of being replaced with random values to improve robustness. Boundary frames are always known and excluded from the loss.

4.5 Facial Expression Generation

As noted in §4.1, HuBERT features encode rich linguistic and prosodic information—phoneme identity, stress, and intonation—which directly drives facial movements: lip shapes follow phonemes, eyebrow raises align with emphasis, smiles correlate with vocal warmth. Sentence-level planning is therefore unnecessary.

We use a Face R-VQVAE (2-layer, codebook size 512, vocabulary of 1,024 IDs) to tokenize 51-dim ARKit blendshape sequences, producing 2 tokens per frame. A Face Infill Transformer, sharing the same architecture and training procedure as the body counterpart (§4.4), generates face tokens directly from audio features. The face pathway runs in parallel with the body pathway at inference time.

Source	# Seqs	Hours	Type
EmbodyAI	84,010	467.7	Conv., Daily
SnapMoGen	20,450	43.7	General
Motion-X	81,084	144.2	Daily, Dance, Sports
Hunyuan Distill.	15,000	20.8	Atomic & Composite
Total	200,544	676.4	—

Table 3: Pre-training data for the Motion Foundation Model. All sequences are retargeted to a unified skeleton and tokenized by the Motion R-VQVAE.

4.6 Motion Foundation Model

To broaden the motion prior beyond conversational data, we pre-train a Motion Foundation Model on 200K+ sequences (676 hours) from diverse sources (Table 3): EmbodyAI (conversations, daily activities), SnapMoGen (general motion), Motion-X (daily, dance, sports), and 15K sequences distilled from the Tencent Hunyuan Motion Model via systematic prompt engineering (details in supplementary materials). All data is retargeted to a unified skeleton and tokenized by the Motion R-VQVAE.

The model is initialized from Qwen-0.5B with motion and audio tokens added to the vocabulary. Pre-training uses a text-to-motion autoregressive objective:

$$P(r_{1:N} | T) = \prod_{i=1}^N P(r_i^1, r_i^2, r_i^3, r_i^4 | T, r_{1:i-1}), \quad (8)$$

where T is a Chinese motion description and N the number of token groups. This gives the downstream planner a broad motion vocabulary and faster convergence on SuSuInterActs.

4.7 Inference Pipeline

At inference, HuBERT extracts audio features from input speech; K-means quantization produces discrete tokens a_i for the LLM,

Table 4: Quantitative comparison on SuSuInterActs. Bold: best; underline: second best. $\uparrow/\downarrow$: higher/lower is better. ESD in seconds. “ $\dagger$ ” indicates T2M-GPT variants with token-by-token autoregressive generation.

Method	Condition	R@1 $\uparrow$	R@2 $\uparrow$	R@3 $\uparrow$	FID $\downarrow$	ESD $\downarrow$	Diversity $\uparrow$
Real Motion	—	62.20	73.56	78.70	0.000	0.308	22.61
<i>Audio-only methods</i>							
EMAGE [30]	Audio	5.00	9.40	13.32	441.6	0.606	12.92
A2M-GPT $\dagger$	Audio	8.72	15.96	20.08	13.66	0.477	22.23
<i>Text-only methods</i>							
HunYuan-Motion	Text	5.21	8.59	11.9	352.56	0.708	16.92
T2M-GPT [65]	Text	23.12	30.49	35.43	67.78	0.721	20.65
MoMask [16]	Text	34.55	46.58	54.29	36.25	0.471	22.03
<i>Audio + Text methods</i>							
AT2M-GPT $\dagger$	Audio, Text	27.52	36.11	41.38	18.491	0.503	22.36
SentiAvatar (Ours)	Audio, Text	43.64	54.94	61.84	8.912	0.456	22.41
Improvement(%)		+26.3	+17.9	+13.9	+34.8	+3.2	+0.2

while continuous features $\mathbf{h}_i$ feed the Infill Transformers. The **body pathway** runs the full two-stage pipeline: the LLM generates sparse keyframe tokens from the motion label $\mathbf{T}$ and audio tokens; the body Infill Transformer then interpolates dense frames via iterative refinement—at each of 6 steps, the most confident predictions are accepted and remaining masked positions continue to the next step; finally, the Motion R-VQVAE decodes tokens into continuous joint rotations. The **face pathway** bypasses the LLM: the Face Infill Transformer generates face tokens from audio features, decoded by the Face R-VQVAE into blendshape coefficients. Both pathways share HuBERT features and run in parallel. For multi-turn dialogue, the LLM’s continuation mode and the Infill Transformer’s sliding window handle cross-utterance transitions. End-to-end latency is ~ 0.3 s for ~ 6 s of output, supporting real-time streaming.

5 Experiments

5.1 Experimental Setup

Datasets. We evaluate our method on two datasets: (1) **SuSuInterActs** (Section 3), our collected single-character Chinese dialogue motion dataset. We split it into 20,982 training, 710 validation, and 542 test samples. (2) **BEATv2** [30], a widely-used English co-speech gesture benchmark. We retrain our full pipeline on the BEATv2 and evaluate on its test split to verify cross-dataset generalization.

Evaluation Metrics. We adopt both objective and subjective metrics. For objective evaluation on the SuSuInterActs dataset, following previous works [16, 65], we report **R@K** ($\uparrow$, $K \in \{1, 2, 3\}$), the text-to-motion retrieval recall measuring semantic alignment; **FID** ($\downarrow$), the Fréchet Inception Distance measuring overall motion quality; **ESD** ($\downarrow$), Event Sync Distance, a bidirectional event-level audio–motion synchronization metric that computes the average nearest temporal distance between detected audio onset events and motion velocity peak events—lower values indicate tighter synchronization (details in the supplementary materials); and **Diversity** ($\uparrow$), the mean pairwise L2 distance among motion latent features. On BEATv2, we follow the standard protocol and report **FGD** ($\downarrow$, Fréchet Gesture Distance), **BC** ($\uparrow$, Beat Consistency), and **Diversity**. For subjective evaluation, we conduct a user study assessing

semantic consistency, prosody synchronization, and overall quality (Section 5.2).

Baselines. We compare with methods spanning three conditioning paradigms. For *audio-only* conditioning, we include **EMAGE** [30], a state-of-the-art co-speech gesture generation method that uses masked audio transformers to synthesize full-body motion from speech audio. For *text-only* conditioning, we include **T2M-GPT** [65], a GPT-based text-to-motion method, and **MoMask** [16], a masked transformer-based text-to-motion method. Since the SuSuInterActs dataset is in Chinese, the original English text encoders used by T2M-GPT and MoMask are not directly applicable. We therefore replace their text encoders with bert-base-chinese for fair comparison, ensuring that these baselines can properly encode our Chinese labels. We also include **HunYuan-Motion** [46], a recent large-scale text-to-motion generation model, as an industry-level zero-shot baseline. For *audio + text* conditioning, we construct two T2M-GPT variants that extend the original text-only GPT architecture with additional input modalities: **A2M-GPT** replaces text with audio tokens as the conditioning signal, while **AT2M-GPT** uses both audio tokens and text labels. Both methods follow the same token-by-token autoregressive generation paradigm as T2M-GPT. For fair comparison, A2M-GPT and AT2M-GPT use the same Qwen-0.5B backbone as our method. These variants help isolate the contribution of our hierarchical two-stage design from the effect of conditioning modalities alone. All baselines (except HunYuan-Motion) are retrained using their official codebases.

Implementation Details. SentiAvatar is trained in three stages on 8×A100 GPUs: (1) *R-VQVAE*: The motion and face R-VQVAEs are trained independently with batch size 128 for 100 epochs. (2) *Motion Foundation Model*: The Qwen-0.5B backbone is pre-trained on the aggregated 200K motion sequences (Table 3) for 10 epochs with per-GPU batch size 128. (3) *SFT*: The pre-trained model is fine-tuned on the SuSu training set for 10 epochs with the same batch size configuration. The Infill Transformers (body and face) are trained with a batch size 1024 for 100 epochs. We use the AdamW optimizer with a learning rate of 1e-4 and a cosine annealing schedule.

Table 5: Comparison on BEATv2. Baseline results from original publications.

Method	Condition	FGD↓	BC↑	Diversity↑
DisCo	Audio	9.417	6.439	9.91
CaMN	Audio, Text	6.644	6.769	10.86
TalkSHOW	Audio	6.209	6.947	13.47
EMAGE	Audio, Text	5.512	7.724	13.06
SynTalker	Audio, Text	6.413	7.971	12.72
Language-of-Motion	Audio	5.301	7.780	15.17
SentiAvatar (Ours)	Audio, Text	4.941	8.078	10.56
Improvement (%)		+6.8	+1.3	–

Table 6: User study results (1–5 Likert scale, higher is better).

Method	Condition	Overall↑	Semantic↑	Prosody↑
HunYuan-Motion	Text	1.88	2.09	2.01
MoMask	Text	2.67	2.79	2.76
EMAGE	Audio	2.48	2.44	2.70
AT2M-GPT	Audio, Text	2.33	2.37	2.45
SentiAvatar (Ours)	Audio, Text	2.99	2.97	3.16
Improvement (%)		+12.0	+6.5	+14.5

5.2 Main Results

Results on SuSuInterActs. Table 4 presents the quantitative comparison. SentiAvatar achieves an R@1 of 43.64%, nearly 2× the best text-only baseline MoMask (34.55%) and 8× EMAGE (5.00%), demonstrating that our LLM planner effectively translates motion labels into semantically consistent keyframe plans. Our FID (8.912) is 4× lower than MoMask (36.25) and 49× lower than EMAGE (441.6), indicating the generated distribution closely matches ground truth. For audio–motion synchronization, SentiAvatar achieves the best ESD (0.456s) among all methods, outperforming EMAGE (0.606s) and text-only methods (T2M-GPT: 0.721s), confirming that the Infill Transformer captures fine-grained speech–motion correspondence. HunYuan-Motion, despite being a large-scale model, scores poorly across all metrics (R@1 5.21%, FID 352.56) as a zero-shot baseline not trained on motion-capture data with our skeleton topology. Diversity (22.41) is comparable to real motion (22.61), indicating the pre-trained foundation model prevents mode collapse.

Results on BEATv2. Table 5 shows that SentiAvatar achieves the best FGD (4.941) and BC (8.078) on this English benchmark, improving the prior best results (FGD 5.301, BC 7.971). This confirms that our architecture generalizes across languages and datasets. The lower diversity (10.56) is expected: our model generates audio-conditioned motion rather than sampling freely from the prior.

User Study. We conduct a user study where 10 participants rate videos on a 1–5 Likert scale across three dimensions: semantic consistency, prosody synchronization, and overall quality. As shown in Table 6, SentiAvatar achieves the highest scores on all three dimensions. The prosody score (3.16) shows the largest margin over baselines, reflecting the benefit of our audio-conditioned Infill Transformer. MoMask achieves the second-best semantic score (2.79) but lags in prosody (2.76) due to its text-only design. EMAGE

Table 7: Architecture ablation on SuSuInterActs.

Variant	R@1↑	FID↓	ESD↓	Div.↑
w/o Pre-training	42.56	8.988	0.452	22.42
w/o Infill Transformer	27.52	18.491	0.503	22.36
w/o LLM Planner	28.06	27.567	0.421	22.33
Full pipeline	43.64	8.912	0.456	<u>22.41</u>

Table 8: Ablation on audio conditioning.

Variant	R@1↑	FID↓	ESD↓	Div.↑
w/o all audio	41.72	9.996	0.523	22.37
w/o audio (Infill)	41.41	10.165	<u>0.497</u>	22.31
w/o audio (LLM)	<u>43.48</u>	<u>9.690</u>	0.517	22.52
Full (both audio)	43.64	8.912	0.456	<u>22.41</u>

Table 9: Ablation on keyframe interval (step t).

Step t	R@1↑	FID↓	ESD↓	Div.↑
Token-by-Token	27.52	18.491	0.503	22.36
$t=2$	34.21	11.773	0.472	22.16
$t=4$	43.64	8.912	<u>0.456</u>	<u>22.41</u>
$t=8$	<u>36.44</u>	<u>9.205</u>	0.439	22.63

scores well on prosody (2.70) but poorly on semantics (2.44), confirming that audio-only methods cannot capture action intent.

5.3 Ablation Study

Architecture ablation. Table 7 validates each component. Removing the LLM planner causes the steepest drop: R@1 falls from 43.64% to 28.06% and FID degrades from 8.912 to 27.567, confirming that sentence-level semantic planning is essential. Its ESD (0.421) is slightly better because the Infill Transformer directly conditions on every audio frame without the keyframe bottleneck, but this local alignment comes at the cost of global semantic coherence. Removing the Infill Transformer degrades all metrics (R@1 27.52%, FID 18.491, ESD 0.503), as sparse keyframes alone produce jerky motion with poor temporal alignment. Pre-training contributes a modest but consistent gain (R@1 42.56% → 43.64%), indicating that the diverse 200K-sequence prior complements task-specific fine-tuning. The full pipeline achieves the best overall balance.

Audio conditioning. Table 8 disentangles the contribution of audio at each stage. Audio in the Infill Transformer is the primary driver of synchronization: removing it worsens ESD from 0.456s to 0.497s, and removing all audio further degrades ESD to 0.523s. Audio in the LLM mainly improves motion quality (FID 9.690 → 8.912) and rhythm planning (ESD 0.517 → 0.456), while text labels carry most semantic content (R@1 drops only 0.16%). Full audio conditioning at both stages yields the best overall performance, confirming the synergy between coarse-grained audio planning and fine-grained audio alignment.

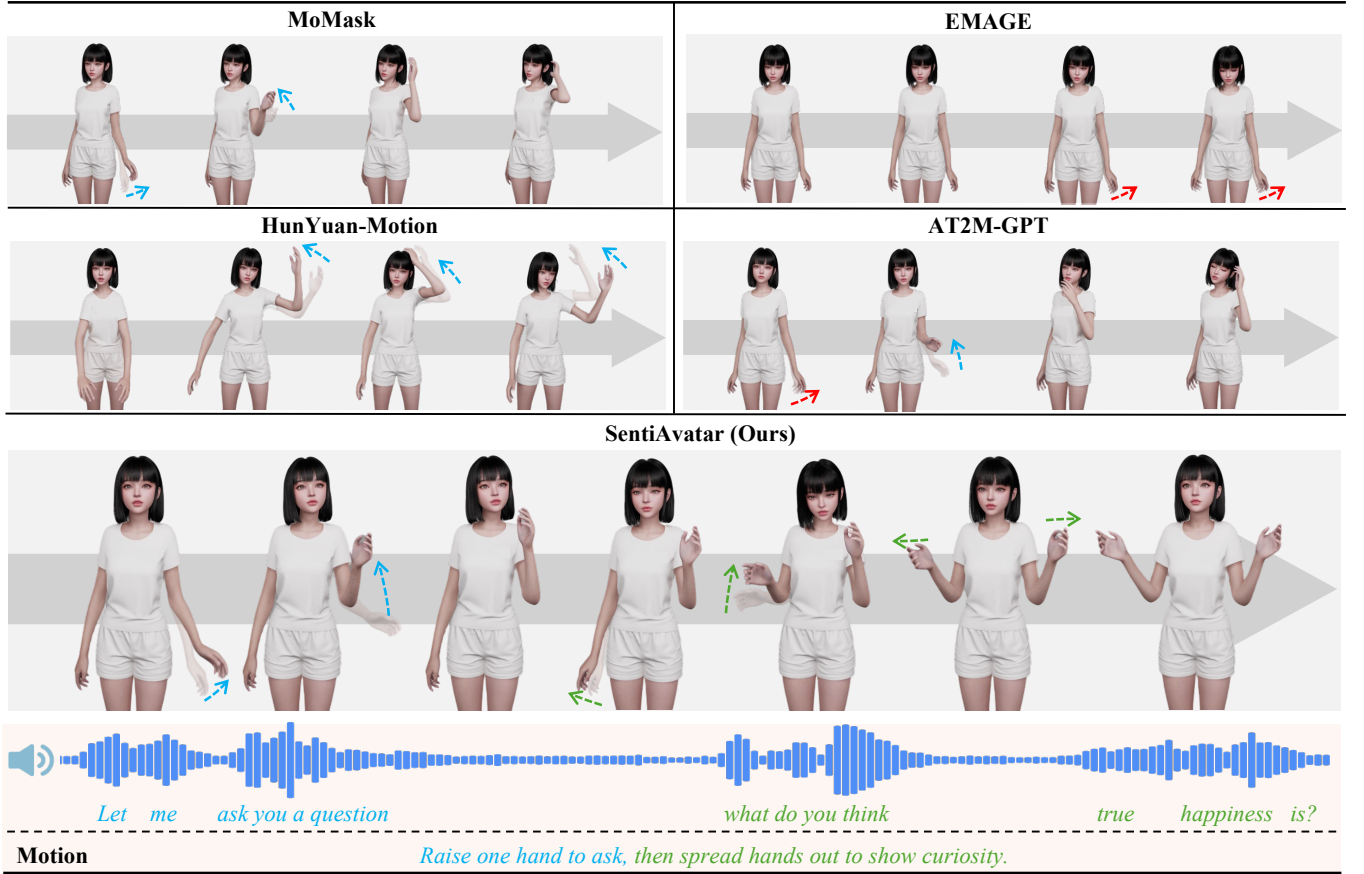


Figure 4: Qualitative comparison of generated motions across methods. Each row shows keyframe sequences for a given motion and speech. Texts and arrows of the same color indicate the same time step. The red arrow indicates an incorrect action.

Keyframe interval. Table 9 studies the step t trade-off. Dense planning ($t=2$) overloads the LLM with long token sequences, degrading R@1 to 34.21%. Sparse planning ($t=8$) achieves the best ESD (0.439s) and diversity (22.63) since the Infill Transformer has more freedom, but semantic alignment drops (R@1 36.44%). The default $t=4$ strikes the best balance (R@1 43.64%, FID 8.912). Token-by-token generation without the Infill Transformer performs worst overall, confirming the advantage of our plan-then-infill design.

5.4 Qualitative Analysis

Figure 4 compares keyframe sequences across methods. **SentiAvatar** produces motions that are both semantically correct (matching the intended action) and temporally aligned with the audio waveform, yielding the most natural results. **MoMask** partially captures action semantics from text labels but generates static-tempo motion with no audio correspondence, as it has no access to speech. **EMAGE** produces audio-synchronized movements—gestures track speech rhythm—but the motions are generic and ignore the semantic intent specified by the label. **AT2M-GPT**, despite receiving both audio and text, often misinterprets the action semantics compared to our hierarchical design, though its motion rhythm roughly follows the audio. **HunYuan-Motion** produces the lowest-quality

output, with visible body distortions and unnatural poses; this is expected since it was not trained on high-quality motion-capture data, further highlighting the value of the SuSuInterActs dataset for character-specific interaction.

6 Conclusion

In this paper, we presented **SentiAvatar**, a novel framework for building highly expressive, interactive 3D digital humans, demonstrated through our real-time character, SuSu. To address the scarcity of interactive conversational data and the challenges of motion-prosody synchronization, we introduced **SuSuInterActs**, a 37-hour multimodal dialogue corpus, and proposed a **plan-then-infill** architecture powered by a pre-trained Motion Foundation Model on over 200K sequences. This design effectively decouples high-level semantic planning from frame-level audio-driven interpolation, ensuring that the generated motions are both contextually accurate and rhythmically aligned with speech. Experiments demonstrate that SentiAvatar achieves state-of-the-art performance on both SuSuInterActs and BEATv2. With highly efficient generation (0.3 s for 6 s of output) and open-sourced resources, our work provides a robust foundation for future research in natural, real-time virtual human interactions.

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